

State dependent hedging

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Abstract

Hedge ratio estimation is a well-researched topic, but the literature remains divided on the economic value of dynamic hedging over a static hedge. We **unify** both strategies under state dependent hedging (SDH), where a hedger's incentive to rebalance depends on the relevant economic state or market condition. The stronger (weaker) the incentive to rebalance, the more dynamic (static) the hedger becomes. Empirically we show that economic policy uncertainty (EPU) in the U.S. affects the statistical properties of NYMEX WTI crude oil in-sample hedge ratios. The out-of-sample relative performance between static and dynamic strategies, depend on which EPU state dominates the evaluation period. SDH utilizes EPU to guide a hedger's rebalancing frequency. It outperforms both static and dynamic strategies, *regardless* of the hedge configuration and econometric model. Static and dynamic hedges are both appropriate, just not all the time, and not at the same time.

JEL classification: G14, G15.

Keywords: Crude oil; Hedging; EPU.

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1 Introduction

Hedge ratio estimation is a major strand of the risk management literature. A large volume of studies address two main issues: First, specify an objective function to derive the corresponding optimal hedge ratio; Second, apply an econometric model to estimate hedge ratio β in-sample, project them out-of-sample, and evaluate against standard benchmarks based on a hedging effectiveness criterion that is consistent with the hedge ratio's objective function.

There are several objective functions to derive an optimal hedge ratio. The most common is the Johnson (1960) and Ederington (1979) risk-minimizing hedge ratio, which minimizes the variance of the hedged portfolio. In order to recognize level increases in the hedged portfolio's payoff, the criterion is generalized to the Howard and DAntonio (1984) risk-return measure, and Kroner and Sultan (1993) expected utility measure¹. Accordingly, the two most common criteria to evaluate hedging effectiveness are risk reduction and Sharpe ratio improvement on the unhedged position.

Many studies utilize a vast range of econometric models to estimate and evaluate hedge ratios for a wide range of markets across asset classes. These can mostly be categorised as OLS, error-correction (ECM), GARCH-type or stochastic volatility (SV) type models. The overall consensus is that, compared to the OLS hedge ratio β_{ols} , ECM, GARCH-type and SV-type hedge ratios provide better hedging effectiveness. Unlike ECM, OLS ignores the spot-futures cointegrating price link², thus omitting a relevant error-correction variable in the β_{ols} estimation; Unlike GARCH-type models³, OLS ignores a conditional covariance matrix that causes the hedge ratio to vary over time β_t ; Unlike SV-type models, β_{ols} does not consider long-memory, mean-reversion, and price (or volatility) jumps. Lastly, β_{ols} cannot be projected out-of-sample for hedging application.

Motivated by the stylized fact that co-movement among markets differ between normal and extreme market conditions, recent studies examine the properties of hedge ratios estimated under different market conditions. Hilal et al. (2011) and Reboredo (2013) incorporate tail dependence to estimate the correlation between a hedging instrument and its target asset. Lien et al. (2016) apply quintile regression to estimate hedge ratios across the spot return distribution. They argue that β_{ols} is averaged across the entire return distribution, and ignores the possibility that it could vary across levels of return. They document an N-shape pattern, with lower hedge ratios observed near the tails of the spot return distribution. Lien et al. (2016) document that a quantile regression hedge ratio outperforms β_{ols} .

Despite voluminous research, the literature does not appear to have reached a consensus on the static versus dynamic hedging debate. This is puzzling given decades of advancement

¹Other objective functions include minimizing the mean extended-Gini (MEG) coefficient in Shalit (1995), generalized semi-variance in De Jong et al. (1997) and Lien and Tse (2000), and VaR of the hedged portfolio in Hung et al. (2006).

²See Ghosh (1993) and Chou et al. (1996) for ECM based hedge ratio estimation.

³Cecchetti et al. (1988) is one of the earliest studies.

in econometric modelling to forecast the conditional covariance matrix between a hedging instrument and its target asset. We make a simple observation; For a given hedge strategy i , if the conditional hedge ratio $h_{i,t}$ is stable over time, the debate is moot. But if $h_{i,t}$ is indeed time-varying, why is the debate unresolved? Our paper’s motivation and contribution is to move the debate towards a consensus. We propose a state-dependent approach that switches the hedge strategy between static and dynamic over time, depending on a pre-specified economic condition or market state that affects the conditional covariance matrix between a hedging instrument and its target asset.

We show that our hybrid hedging strategy outperforms both static and dynamic hedging, regardless of the actual hedging configuration e.g. estimation window, sample period, re-balancing frequency, econometric model, hedging instrument and target asset. Hedging results that do not depend on the hedging configuration, are crucial to promote consensus in the literature. Consider 5 econometric models and 5 different out-of-sample periods for a given hedging instrument and target asset. This could lead to 25 sets of possibly mixed results on static versus dynamic hedging. Further considering 5 asset-types will expand to a possible 125 sets of results.

Our proposed hybrid strategy is implemented with the same configuration as static and dynamic hedging. This provides a convincing analysis on the economic significance of state-dependent hedging, as a pathway towards consensus. We view a dynamic hedge as a static hedge with an option to re-balance the hedging position, the value of which is expected to increase with the volatility in $h_{i,t}$. To follow, a stable $h_{i,t}$ would trivialize the re-balancing choice. For a hedging instrument and target asset, changes in the conditional covariance matrix can lead to $h_{i,t}$ becoming either stable or volatile over time. If $h_{i,t}$ is generated from different econometric models, for different out-of-sample evaluation period, and across different asset markets, this would generate a dispersion of empirical findings across a large volume of studies on the incremental benefits of a dynamic hedge strategy⁴.

State-dependent hedging incorporates an economic state (or market condition) to guide the re-balancing choice. Ex-ante, if a hedger is aware of an economic state that is associated with $h_{i,t}$ becoming volatile (stable), during the out-of-sample period when the state occurs, this becomes a signal to switch to dynamic (static) hedging. Our hybrid approach stipulates that static and dynamic hedging are both appropriate, just not all the time, and not at the same time. As to what is a relevant economic state or market condition, this is an empirical question that depends on the asset market and econometric model which is used to generate $h_{i,t}$. More importantly, the concept of state-dependent hedging is compatible with any asset market or econometric model.

Our empirical application is based on the West Texas Intermediary (WTI) crude oil market. It is the most important commodity market based on turnover volume and first-

⁴This could even be possible when two published papers use the same econometric model on the same asset markets, but for different sample periods. If starkly different economic states underlay the two evaluation periods, it is possible that the two papers report opposing findings on the benefits of dynamic hedging.

order impact on major sectors of the global economy. Unlike agricultural commodities, crude oil hedging could offer an interesting dimension in that there is non-trivial hedging demand from both producing and consuming industries. Given that government economic policy and crude oil are related to the overall U.S. economy, we conjecture that the crude oil market is affected by economic policy uncertainty (EPU), which we proxy using the Baker, Bloom and Davis (BBD 2016) EPU_t index. To note, we are not advocating EPU_t to be the most appropriate economic state variable to explain crude oil hedge ratios variations over time. State-dependent hedging only requires a hedger to identify a relevant economic state or market condition that is associated with $h_{i,t}$ becoming either stable or volatile over time. Empirically, if EPU_t is not an appropriate state variable for crude oil $h_{i,t}$, this would make our positive empirical findings more convincing.

SUMMARY OF MAIN FINDINGS and IMPLICATIONS

The next section outlines our sampling and institutional details. Section 3 presents the in-sample estimation results, while Section 4 reports out-of-sample hedging evaluation. Section 5 concludes.

All countries consume crude oil or oil products. Both producers and consumers are highly concerned about crude oil prices.

The crude oil prices are being directly affected by several economic, political, geopolitical, technological factors, and also oil reserves, available stocks and weather conditions, among others.

For the modern economy, no commodity is arguably more important than crude oil. This is true in terms of the effects of crude oil pricing on real economic activity, as well as financial activity in the trading of crude-oil related securities.

On other hand the crude oil price fluctuations influence directly the world economy. Economic agents indirectly involved in crude oil negotiations, such as firm or government planners, are looking for related petroleum price forecasting models, elaborating studies, while the agents directly involved are looking for the hedge strategies studies as well.

Consequently these prices are determinant for hedge strategies studies. The hedge strategies allow negotiators that have short and long positions in the market protection against prices fluctuations.

Affected by many factors like production, import, export and economic fundamentals, most of commodities face tremendous fluctuations of price generally, which makes it crucially important to manage price risks.

Take crude oil market as an example, sharp fluctuations in crude oil prices will impact inflation levels and the economic development of various countries around the world.

In 2019, intensified trade disputes and drastic geopolitical changes triggered several shocks in the oil market.

In 2020, the COVID-19 epidemic spread over the world, coupled with the complexity of the international situation, the international oil price collapsed.

On April 21st, 2020, WTI crude oil futures plunged 300% to 37.63 dollars/barrel. Global crude oil market experienced A night of terror in the history of crude oil trading. This round of sharp drop in oil prices has brought a huge impact on the crude oil production industry and energy enterprises.

Affected by the sharp drop in the price of crude oil futures contracts in the United States, the three major stock indexes in the United States continued to fall.

Among them, the Dow Jones Industrial Average closed at 23018.88 points, down 631.56 points, or 2.67%. The S P 500 index and the NASDAQ index fell 3.07% and 3.48% respectively. The Dow Jones Industrial Average has lost more than 1200 points in two trading days.

Johnson (1960), Ederington (1979), Myers and Thompson (1989) used the minimum-variance hedge model to obtain the hedge ratio.

Feng et al. (2012), Reboredo and Ugando (2015), Segnon et al. (2017), Abadie et al. (2017), Yu et al. (2018) studied the hedging problems based on minimum-VaR models.

Tekiner-Mogulkoc et al. (2015), Lu et al. (2016), Hemmati et al. (2016), Liu et al. (2017), Roustai et al. (2018), Chai and Zhou (2018) decided the hedge ratio by using the objective of minimum-CVaR.

Most of existing literature argue that the asset return follows a normal distribution, however, the actual return distribution of assets has more leptokurtosis and fatter tails than the normal distribution (Naqvi et al.,2017, Bara nano et al., 2020), so its likely that the normal assumption would lead up to the underestimate of risks and the bias of hedging strategy to a large extent.

In light of this, this paper investigates the effect of skewness on hedging decision through numerical analysis and empirical test to WTI crude oil and Brent crude oil.

Crude oil is a kind of important commodity and strategic materials, **correlated with economic and financial activities tightly** (Bhutto et al., 2020; Falkowski et al.,2020; Hamiton, 2009; Wang et al.,2019).

Then, out of its own characteristics, crude oil has the relatively bigger volatility and skewness, which offers great research foundation.

Finally, from the perspective of existing literature, research based on the background of crude oil does not account for the skewness risks in most cases, which may bring the problem of risk mis-estimation.

2 Sampling and institutional background

2.1 Conceptual framework

We present a simple conceptual framework to motivate the economic value of state-dependent hedging (SDH), which comes from timing the benefits of static and dynamic hedge associated with different economic states or market conditions.

Consider a short-hedger in equation (1), whose hedged portfolio return $r_{p,t}$ at time t is the difference between the return from a unit spot position $r_{s,t}$, and futures return $r_{f,t}$ from h_{t-1} unit of short position that is established at $t - 1$. Taking the variance of $r_{p,t}$ gives $\sigma_{p,t}^2$. The hedger has a mean-variance utility function with risk-aversion parameter $\gamma > 0$. At each time t , the hedger solves an optimal h_t^* to maximize the expected utility of next period's hedged portfolio return $E_t U(r_{p,t+1})$. The solution h_t^* evokes a trade-off between the expected benefit of risk-reduction through $\sigma_{sf,t+1}$, and the expected return $E_t(r_{f,t+1})$ from the futures position. Both terms are normalized by the futures variance $\sigma_{f,t+1}^2$.

$$\begin{aligned}
 r_{p,t} &= r_{s,t} - h_{t-1} r_{f,t} \\
 \sigma_{p,t}^2 &= \sigma_{s,t}^2 + h_{t-1}^2 \sigma_{f,t}^2 - 2h_{t-1} \sigma_{sf,t} \\
 E_t U(r_{p,t+1}) &= E_t(r_{p,t+1}) - \gamma \text{Var}_t(r_{p,t+1}) \\
 \max_{h_t} E_t U(r_{p,t+1}) &= \max_{h_t} [E_t(r_{s,t+1}) - h_t E_t(r_{f,t+1}) - \gamma(\sigma_{s,t+1}^2 + h_t^2 \sigma_{f,t+1}^2 - 2h_t \sigma_{sf,t+1})] \\
 h_t^* &= \frac{-E_t(r_{f,t+1})}{2\gamma \sigma_{f,t+1}^2} + \frac{\sigma_{sf,t+1}}{\sigma_{f,t+1}^2}
 \end{aligned} \tag{1}$$

Kroner and Sultan (1993) show that if futures price f_t follows a martingale $E_t[f_{t+1}] = f_t$, h_t^* simplifies to a risk-minimizing conditional hedge ratio $\frac{\sigma_{sf,t+1}}{\sigma_{f,t+1}^2}$. Intuitively, if $E_t(r_{f,t+1}) = 0$, a mean-variance hedger would solve h_t^* , focusing only on risk reduction. But if $E_t(r_{f,t+1}) > 0$ i.e. sub-martingale, the hedger would trim back h_t^* in response to an expected loss at $t + 1$ from a short hedge. Conversely, if $E_t(r_{f,t+1}) < 0$ i.e. super-martingale, the same short hedger expects a short hedge to profit at $t + 1$, and solves h_t^* to be higher than what is guided by risk-minimization. The trade-off is also affected by the risk aversion parameter γ ; An extremely risk-averse hedger (high γ) would focus predominately on the risk-minimizing term. But for a moderately risk-averse hedger (low γ), both risk reduction and return enhancement matter in deciding the futures position.

A static hedger has the right, but not the obligation to rebalance a hedge position. We view a dynamic hedge as a static hedge plus a series of rebalancing options over the hedging

period. It is reasonable to assume that a hedger's incentive to rebalance is closely related to the performance evaluation criterion, as well as transaction cost. The latter includes trading fees, brokerage commission, and capital commitment associated with a margin account.

$$\text{Dynamic hedge} = \text{Static hedge} + \sum_t^T \text{Rebalancing option}$$

$$\text{Value(Rebalancing option)} = f[\text{Performance criterion; Transaction cost, Margin commitment}]$$

Compare a conditional hedge ratio h_t against the hedge ratio h_x that is used to strike the current futures position. A large $[h_t - h_x] > 0$ must imply a large $\sigma_{sf,t}$, relative to $\sigma_{f,t}^2$. In equation (1), $\sigma_{p,t}^2$ shows that a hedger's incentive to rebalance, increases with $[h_t - h_x]$. When a hedger exercises the option to rebalance, such that $h_x \rightarrow h_t$, both $h_t^2 \sigma_f^2$ and $2h_t \sigma_{sf}$ increase. But the latter has a greater magnitude, leading to a net risk-reduction in $\sigma_{p,t}^2$. Conversely, if h_t decreases substantially after a hedge position has been struck on h_x , this must correspond to $\sigma_{sf,t}$ becoming small, relative to $\sigma_{f,t}^2$. The same hedger's incentive to reduce the futures position, increases with $[h_x - h_t] > 0$. When a hedger exercises the option to reduce the futures position i.e. $h_x \rightarrow h_t$, $h_t^2 \sigma_f^2$ decreases by a greater magnitude than $2h_t \sigma_{sf}$, resulting in a net risk-reduction in $\sigma_{p,t}^2$.

We add more conceptual structure by deriving a mean-variance hedger's rebalancing incentive from equation (1). At time t , a short hedger updates $h_t|I_t$, which is the hedge ratio conditional on the hedger's information set I_t . The hedger has an incentive to increase a short position if the expected utility of next period's hedged portfolio return given the updated hedge ratio h_t i.e. $E_t U(r_{p,t+1}|h_t)$, exceeds that which is based on an upper threshold hedge ratio $E_t U(r_{p,t+1}|h_t^u)$. To follow, the short hedger has an incentive to decrease a short position if $E_t U(r_{p,t+1}|h_t) > E_t U(r_{p,t+1}|h_t^l)$. Later below, we elaborate on the conceptual meaning of (h_t^u, h_t^l) , and their empirical implication for state-dependent hedging.

INSERT FIGURE X: HEDGE RATIO UPDATING

Figure X illustrates the implementation of state-dependent hedging over time. At $t = 0$, an initial hedge is established based on $h_0 \in [h_0^u - h_0^l]$, where $h_0^u = h_0 + \sigma_{h_t}$, $h_0^l = h_0 - \sigma_{h_t}$, and σ_{h_t} is the volatility of h_t . For a static hedger, $h_0 = h_1 = \dots = h_T$, where T is the end of the hedging period. For a dynamic hedger, h_t varies over $t = 1, 2, \dots, T$, and rebalancing occurs whenever fluctuation in h_t induces a change in the number of futures contracts. At $t = 1$, a state-dependent hedger estimates a conditional hedge ratio $h_1|I_1$. If $h_1 \in [h_0^u - h_0^l]$, no rebalancing occurs, such that $h_1 = h_0$, $h_1^u = h_0^u$ and $h_1^l = h_0^l$. But if $h_1 \notin [h_0^u - h_0^l]$, the hedger rebalances and updates $h_0 \rightarrow h_1 \in [h_1^u - h_1^l]$, where $h_1^u = h_1 + \sigma_{h_t}$ and $h_1^l = h_1 - \sigma_{h_t}$. The same applies at $t = 2$; If $h_2 \in [h_1^u - h_1^l]$, then $h_2 = h_1$, $h_2^u = h_1^u$ and $h_2^l = h_1^l$; Otherwise, the hedger updates $h_1 \rightarrow h_2 \in [h_2^u - h_2^l]$, where $h_2^u = h_2 + \sigma_{h_t}$ and $h_2^l = h_2 - \sigma_{h_t}$, and so forth. Denote tc as the cost to rebalance, which includes transaction cost, brokerage fees and changes to committed capital to maintain the margin account.

To increase short hedge:

$$\begin{aligned}
& E_t U(r_{p,t+1}|h_t) - tc > E_t U(r_{p,t+1}|h_t^u) \\
& - tc - E_t(r_{f,t+1})h_t - \gamma(\sigma_{s,t+1}^2 + h_t^2\sigma_{f,t+1}^2 - 2h_t\sigma_{sf,t+1}) > \\
& - E_t(r_{f,t+1})h_t^u - \gamma(\sigma_{s,t+1}^2 + h_t^{u2}\sigma_{f,t+1}^2 - 2h_t^u\sigma_{sf,t+1}) \\
& \gamma(2\sigma_{sf,t+1}[h_t - h_t^u] - \sigma_{f,t+1}^2[h_t^2 - h_t^{u2}]) - E_t(r_{f,t+1})[h_t - h_t^u] > tc \\
& \gamma[h_t - h_t^u] \left(2\sigma_{sf,t+1} - \sigma_{f,t+1}^2[h_t + h_t^u] - \frac{E_t(r_{f,t+1})}{\gamma} \right) > tc \tag{2}
\end{aligned}$$

In equation (2), a hedger increases a short position when all 3 conditions are met:

1. $[h_t - h_t^u] > 0$
2. $\gamma(2\sigma_{sf,t+1} - \sigma_{f,t+1}^2[h_t + h_t^u]) > E_t(r_{f,t+1})$ i.e. A large conditional spot-futures covariance $\sigma_{sf,t+1}$ relative to the futures conditional variance $\sigma_{f,t+1}^2$ generates a variance reduction benefit, which is required to be greater than $E_t(r_{f,t+1})$.
3. The values associated with 1. and 2. has to exceed transaction cost tc .

A short hedger's incentive to increase a futures position is triggered by Condition 1 $[h_t - h_t^u] > 0$, which resembles an option's payoff. Due to transaction cost tc , Condition 1 is necessary, but not sufficient. Rebalancing depends on two other conditions. Condition 2 relates to the expected benefit in variance reduction; Increasing a futures position loads $\sigma_{f,t+1}^2$ onto the hedged portfolio, but reduces risk through return covariance with the spot position. When $(2\sigma_{sf,t+1} - \sigma_{f,t+1}^2[h_t + h_t^u])$ is large, increasing a short hedge translates to a larger expected benefit in variance reduction. Condition 3 states that the expected benefit from rebalancing has to exceed the associated transaction costs tc .

If futures price F_t follows a martingale $E_t(r_{f,t+1}) = 0$, the hedger compares the expected benefit of variance reduction against tc . But if F_t follows a sub-martingale $E_t(r_{f,t+1}) > 0$, Condition 2 shows that the expected utility gain from risk reduction, first has to overcome the expected loss from increasing a short hedge. For a moderately risk-averse (low γ) hedger to rebalance, a large risk reduction is required to compensate the expected loss from the futures position. Conversely, an extremely risk-averse (high γ) hedger gains a large expected utility from risk reduction, and is likely to increase a short hedge, in spite of the expected loss. Lastly, if F_t follows a super-martingale $E_t(r_{f,t+1}) < 0$, the hedger expects to profit from increasing a short hedge. This complements the risk reduction benefit, giving the hedger extra incentive to rebalance. A high (low) γ hedger draws more utility from risk reduction (return enhancement) associated with increasing short hedge.

To decrease short hedge:

$$\begin{aligned}
& E_t U(r_{p,t+1}|h_t) > E_t U(r_{p,t+1}|h_t^l) \\
& - tc - E_t(r_{f,t+1})h_t - \gamma(\sigma_{s,t+1}^2 + h_t^2\sigma_{f,t+1}^2 - 2h_t\sigma_{sf,t+1}) > \\
& - E_t(r_{f,t+1})h_t^l - \gamma(\sigma_{s,t+1}^2 + h_t^{l2}\sigma_{f,t+1}^2 - 2h_t^l\sigma_{sf,t+1}) \\
& \gamma(\sigma_{f,t+1}^2[h_t^{l2} - h_t^2] - 2\sigma_{sf,t+1}[h_t^l - h_t]) + E_t(r_{f,t+1})[h_t^l - h_t] > tc \\
& \gamma[h_t^l - h_t] \left(\sigma_{f,t+1}^2[h_t^l + h_t] - 2\sigma_{sf,t+1} + \frac{E_t(r_{f,t+1})}{\gamma} \right) > tc \tag{3}
\end{aligned}$$

Equation (3) outlines a short hedger's incentive to decrease a futures position, which depends on a similar set of conditions as those from equation (2). First, h_t has to decrease below a lower threshold h_t^l . The second is $(\sigma_{f,t+1}^2[h_t^l + h_t] - 2\sigma_{sf,t+1}) > 0$, which indicates an expected net risk reduction in the hedged portfolio. When $(\sigma_{f,t+1}^2)$ is large relative to $\sigma_{sf,t+1}$, reducing a short position offloads more risk through $\sigma_{f,t+1}^2$, than the reduced risk cancellation through $\sigma_{sf,t+1}$. When $E_t(r_{f,t+1}) = 0$, the risk reduction benefit is compared directly against tc . If $E_t(r_{f,t+1}) > 0$, the short hedger has more incentive to decrease a short hedge, since it helps reduce the expected loss. But if $E_t(r_{f,t+1}) < 0$, reducing a short hedge decreases the expected gain, giving the hedger less incentive to rebalance. The trade-off between expected profit and expected risk reduction also depends on the hedger's risk-aversion γ .

To note, we do not intend for equations (2) and (3) to solve a mean-variance hedger's optimal rebalancing conditions. Our main purpose is only to show that a hedger's incentive to rebalance, depends on $[h_t - h_t^u] > 0$ and $[h_t^l - h_t] > 0$, as well as risk and return components that are found in commonly used hedge performance measures in the literature. Starting with expected utility payoffs from rebalancing, we derive the necessary conditions $[h_t - h_t^u] > 0$ and $[h_t^l - h_t] > 0$. This shows that $E_t[U(r_{p,t+1})]$, as well as the value of a static hedger's rebalancing option, are monotonic increasing in the volatility of the conditional hedge ratio σ_{h_t} . If h_t becomes significantly more volatile during a state of high EPU, this increases the probability of occurrence for $[h_t - h_t^u] > 0$ and $[h_t^l - h_t] > 0$. The expected net benefit from rebalancing, hence the relative performance of static versus dynamic hedges, depends on which EPU state dominates a given out-of-sample period. In SDH, the rebalancing frequency varies across different EPU states, which drives its economic value over static and dynamic hedge strategies.

In our conceptual framework, h_t^u and h_t^l are functions of σ_{h_t} . If rebalancing occur at time $t - 1$, h_t^u is updated to $h_{t-1} \pm \sigma_{h_t}$; Otherwise, $h_t^u = h_{t-1}^u$. Hence, $h_t^u = [h_{t-1}^u; h_{t-1} + \sigma_{h_t}]$, and $h_t^l = [h_{t-1}^l; h_{t-1} - \sigma_{h_t}]$. At time t , a hedger estimates $h_t|I_t$ and checks if $h_t \in [h_t^u - h_t^l]$. The rebalancing condition $[h_t - h_t^u] > 0$ implies $\Delta h_t > \sigma_{h_t}$. To rebalance, the increase from h_{t-1} (which sets the futures position at time t) to h_t has to be sufficiently large that it exceeds σ_{h_t} . The futures position is recalculated on h_t , and the short position is increased accordingly.

Similarly, $[h_t^l - h_t] > 0$ implies $-\Delta h_t > \sigma_{h_t}$; To reduce a short position would require the decrease in h_t to exceed σ_{h_t} .

In sum, Condition 1 from equations (2) and (3) can be summarized as $|\Delta h_t| \geq \sigma_{h_t}$. A static hedger does not update the hedge ratio $|\Delta h_t| = 0$; A dynamic hedger regularly updates the hedge ratio $|\Delta h_t| > 0$; In contrast, a state-dependent hedger would only update h_t to rebalance, if the change is non-trivial $|\Delta h_t| - \sigma_{h_t} > 0$. Otherwise, the hedger does not rebalance $|\Delta h_t| = 0$. Under SDH, rebalancing takes the form $\text{Max}(|\Delta h_t| - \sigma_{h_t}, 0)$. When we compare against rebalancing under static and dynamic strategies, it draws out the rebalancing option mechanism of SDH.

Static hedger : $|\Delta h_t| = 0$

Dynamic hedger : $|\Delta h_t| > 0$

State-dependent hedger : $|\Delta h_t| = \text{Max}(|\Delta h_t| - \sigma_{h_t}, 0)$

To note, σ_{h_t} is estimated from in-sample hedge ratios. If the out-of-sample hedge ratio $h_{oos,t}$ exhibits significantly different volatility $\sigma_{t,h_{oos,t}}$ under different EPU states, this directly affects a hedger's incentive to rebalance. For example, during a high EPU state, if $\sigma_{t,h_{oos,t}}$ is significantly higher than σ_{h_t} , it is more likely for $|\Delta h_{oos,t}| - \sigma_{h_t} > 0$ to occur. Accordingly, rebalance frequency is expected to increase during a high EPU state, such that SDH resembles a dynamic strategy. Conversely, assume a low EPU state is associated with less volatile hedge ratios, such that $\sigma_{t,h_{oos,t}}$ is significantly lower than σ_{h_t} . Now it is less likely for $|\Delta h_{oos,t}| - \sigma_{h_t} > 0$ to occur, such that SDH resembles a static hedge.

The rebalancing condition $|\Delta h_t| > \sigma_{h_t}$, and more generally SDH, has **three noteworthy appeals**, especially for practitioners. **First**, the implementation of SDH is not econometric model-specific. As the empirical section will show, a hedger can use model $i=\{\text{OLS, VEC, DCC, SV}\}$ to generate in-sample h_t , estimate σ_{h_t} and implement SDH out-of-sample. **Second**, SDH is not an asset-specific concept that is limited to crude oil hedging. The concept applies to any asset whose spot-futures conditional covariance matrix is affected by a given economic state or market condition. **Third**, our conceptual framework is based on a mean-variance utility function, which accommodates a risk-minimizing hedger. If futures prices follow martingale, or if risk aversion γ is high, a mean-variance hedger becomes a risk-minimizing hedger.

Indeed, EPU-based SDH focuses on risk reduction. When a hedge ratio turns volatile, the risk reduction trade-off between $\sigma_{sf,t}$ and $\sigma_{f,t}^2$ becomes unstable. When this happens, more rebalancing is required to ensure the futures position delivers a net risk-reduction to the hedged portfolio. For risk-based performance criteria e.g. Edrington's risk-reduction measure, EPU-based SDH should outperform both static and dynamic hedges. More importantly, SDH can accommodate risk-reward performance criteria e.g. Sharpe Ratio. A mean-variance hedger's incentive to rebalance now also depends on $E_t(r_{f,t+1})$. This suggests the economic

value of SDH could be enhanced by a second market state variable to indicate whether the crude oil futures market is in backwardation, martingale (neutral) or contango. In this ‘bivariate’ SDH, market state guides rebalancing on return enhancement, while EPU state guides rebalancing on risk reduction. The next section elaborates.

2.2 Bivariate SDH: Economic state and market state

According to Cootner (1960) and Hirshleifer (1988), unlike financial assets, commodity markets experience net hedging pressure, and futures prices do not tend to follow martingale. Agricultural markets are typically described by large producers dealing with smaller commercial users. This creates a pervasive net hedging pressure on the short side. To attract rent-seeking long speculators and investors, farmers have to leave $E_t(r_{f,t+1}) > 0$ on the hedging table. This explains why agriculture futures prices tend to follow sub-martingale $E_t[F_{t+1}] > F_t$. Conversely, if a commodity market experiences net hedging pressure on the long side, $E_t(r_{f,t+1}) < 0$ is required to attract short speculators and investors. For these commodities, F_t tends to follow a super-martingale $E_t[F_{t+1}] < F_t$.

The futures pricing literature relates the above closely to the market states of backwardation $F_t < E_t(S_T)$ and contango $F_t > E_t(S_T)$, where $E_t(S_T)$ is the expected spot price at the futures contract maturity date T . During backwardation (contango), F_t is expected to drift up (down) over time, which corresponds to a sub (super) martingale. But unlike many commodities, the crude oil market is populated by large producers as well as commercial users. Since production and consumption cycles often do not synchronise, net hedging pressure is a stochastic process. This partly explains why the crude oil market oscillates more frequently between backwardation and contango, compare to other commodities.

The presence of $E_t(r_{f,t+1})$ in equations (2) and (3) suggests that SDH could benefit from a second market state variable, denote as BMC_t , to indicate whether the WTI crude oil futures market is in backwardation, martingale or contango. This affects a mean-variance hedger’s rebalancing incentive through return enhancement. If BMC_t and EPU_t are correlated, it further justifies utilizing EPU_t to guide state-dependent hedging. But if BMC_t and EPU_t are weakly correlated, the economic value of SDH could be enhanced by utilizing BMC_t to guide rebalancing on return enhancement, while EPU_t guides rebalancing on risk reduction. The parameter γ affects a hedger’s emphasis between the two benefits. We provide a brief conceptual outline, and some empirical results to demonstrate the implementation. The full discussion and formal empirical analysis on bivariate SDH is presented in a separate paper.

We extend the conceptual framework of EPU-based SDH, following the implementation in Figure X. From Condition 1: $[h_t - h_t^u] > 0$, rebalancing takes the form $\text{Max}(|\Delta h_t| - \sigma_{h_t}, 0)$, where σ_{h_t} is the volatility of in-sample hedge ratio. In univariate SDH, we use EPU_t to form in-sample h_t sub-samples for different EPU states, and test for significant differences in σ_{h_t} . We confirm that h_t is more volatile during a state of high EPU. If this statistical property holds out-of-sample i.e. $\sigma_{h_{oos,t}}^h > \sigma_{h_t}$, then it is more likely for $|h_{oos,t}| - \sigma_{h_t} > 0$ to occur. As

such, SDH resembles a dynamic (static) strategy during a high (normal/low) EPU state.

In bivariate SDH, the interaction between BMC and EPU generates four implementation modes that represent a rebalancing frequency spectrum; Static mode $[-,-]$ and dynamic mode $[+,+]$ sit at the ends of the spectrum, while quasi-static $[+,-]$ and quasi-dynamic $[-,+]$ modes correspond to when BMC state and EPU state guides rebalancing.

1. **Static mode** $[-,-]$: Consider a low/normal EPU state, where $\sigma_{t,h_{oos,t}} = \sigma_{h_t}$, and the futures market is in martingale $E_t(r_{f,t+1}) = 0$. In this scenario, it is unlikely for $|h_{oos,t} - \sigma_{h_t}| > 0$ to frequently occur, such that SDH resembles a static hedge.
2. **Quasi-static mode** $[+,-]$: Assume the economy remains in a low/normal EPU state $\sigma_{t,h_{oos,t}} = \sigma_{h_t}$, but the futures market shifts to either backwardation or contango $E_t(r_{f,t+1}) \neq 0$. The transition to a new market state triggers an initial rebalancing, but it does not affect subsequent rebalancing incentive.
3. **Quasi-dynamic mode** $[-,+]$: Assume the economy enters a high EPU state $\sigma_{h_{oos,t}}^h > \sigma_{h_t}$, but the futures market is in martingale. As $\frac{-E_t(r_{f,t+1})}{2\gamma\sigma_{f,t+1}^2} = 0$, a mean-variance hedger focuses on risk-minimization; Bivariate SDH reverts to EPU-based SDH.
4. **Dynamic mode** $[+,+]$: Now assume the futures market shifts away from martingale $E_t(r_{f,t+1}) \neq 0$, and the economy enters a high EPU state $\sigma_{h_{oos,t}}^h > \sigma_{h_t}$. Here, SDH resembles a dynamic hedge.
 - Contango $E_t(r_{f,t+1}) < 0$: A hedger computes $h_{oos,t}$ with $\frac{-E_t(r_{f,t+1})}{2\gamma\sigma_{f,t+1}^2} > 0$. In conjunction with $\sigma_{h_{oos,t}}^h > \sigma_{h_t}$, it is more likely for $\Delta h_{oos,t} - \sigma_{h_t} > 0$ i.e. increase short hedge or decrease long hedge, compare to univariate SDH.
 - Backwardation $E_t(r_{f,t+1}) > 0$: A hedger computes $h_{oos,t}$ with $\frac{-E_t(r_{f,t+1})}{2\gamma\sigma_{f,t+1}^2} < 0$; More likely for $-\Delta h_{oos,t} - \sigma_{h_t} > 0$ i.e decrease short hedge or increase long hedge, compare to univariate SDH.

In quasi-static mode, a change in market state affects the level of $h_{oos,t}$, but not its volatility. Unlike EPU, a change in BMC state delivers a once-off effect on rebalancing. When a futures market transits from martingale to contango, $h_{oos,t}$ increases; The hedger is likely to increase a short hedge $\Delta h_{oos,t} > \sigma_{h_t}$ with an intention to profit from $E_t(r_{f,t+1}) < 0$. Conversely, when the market switches from martingale to backwardation, the short hedger computes a smaller $h_{oos,t}$, and is likely to trim back a short hedge to avoid losses associated with $E_t(r_{f,t+1}) > 0$. In both cases, after an initial rebalancing, the hedger updates (h_t^u, h_t^l) based on $h_{oos,t-1}$, which has been market-state adjusted. As such, the hedger has no further incentive to rebalance, while remaining in the same market state.

The quasi-dynamic mode is equivalent to EPU-based SDH; Since $E_t(r_{f,t+1}) = 0$, a mean-variance hedger focuses on risk-minimization. When $h_{oos,t}$ becomes volatile, the trade-off between $\sigma_{sf,t+1}$ and $\sigma_{f,t+1}^2$ is unstable. This increases the incentive to rebalance in order to manage the trade-off over time in order to maintain a net risk-reduction benefit to minimize risk in the hedged portfolio. Lastly, in dynamic mode, BMC_t and EPU_t alter the level and volatility of $h_{oos,t}$, which makes rebalancing more frequent.

2.3 Institutional background

Copy Paste Institutional details from thesis; The US Department of Energy Energy Info Admin (EIA) has plenty of details, charts and some data on crude oil.

OHR estimation (Cecchetti et al. (1988); Myers and Thompson (1989); Baillie and Myers (1991); Kroner and Sultan (1991); (Lien and Luo 1993); Lin et al. (1994); Strong and Dickinson (1994); Park and Switzer (1995).

The general consensus is that the use of multivariate generalized GARCH models yields superior performances, evidenced by lower portfolio volatilities, than either time-invariant or rolling OLS hedges.

Cecchetti et al. (1988), Myers and Thompson (1989), Baillie and Myers (1991) and Kroner and Sultan (1991) argue that commodity prices are characterized by time-varying covariance matrices. As news about spot and futures prices arrives to the market, the conditional covariance matrix and, hence, the optimal hedging ratio become time-varying. They employ MGARCH models to capture time variation in the covariance matrix and the resulting hedge ratio.

On the other hand, there is also evidence that the benefits of a time-varying hedge are substantially diminished as the duration of the hedge is increased (e.g., Lin et al. 1994).

1. we link the concept of the optimal hedge with Kroner and Ngs(1998) notion of the news impact surface. The hedging surface of the OLS model is independent of news arriving to the market and therefore could be suboptimal.

2. We extend the models of Cecchetti et al.(1988), Myers and Thompson (1989), and Baillie and Myers (1991) to allow for time variation and asymmetry across the entire variance-covariance matrix of returns. This means that the hedge ratio will be sensitive to the size and sign of the change in prices resulting from information arrival.

Details of NYMEX WTI crude futures.

2.4 Data and sampling

Our daily sample runs from 1995 to 2020. A hedging paper usually pre-specifies the in-sample and out-of-sample partitions to evaluate hedge ratios. As this paper focuses on a hybrid approach, where a hedger switches between static and dynamic hedging, there is no universal in-sample period that applies to all the analyses. We highlight a crucial step for state-dependent hedging. For a chosen state variable, in this case EPU_t , we need to identify thresholds that can indicate when the state of EPU has shifted. While actual changes in EPU states are unobservable, we

run in-sample tests on sub-samples that are formed based on EPU_t breakpoints, and solve for the implied thresholds that generate the most significant test statistics on differences in sub-samples between (say) the High and Low EPU states. That in-sample period cannot be used for any out-of-sample evaluation. More generally, we are careful not to instill forward-looking biases into our out-of-sample hedging evaluation; Any model estimation period cannot cross into the out-of-sample evaluation period.

As commodities are traded in multiple venues around the world, there is no centralized spot market to speak of. Following standard convention, we use the NYMEX West Texas Intermediate (WTI) front contract as a proxy for the spot market, and construct spot variables using data from the front contract. To follow, we designate the second month contract as the futures market. In our preliminary analysis, we compare daily open interest between the first two contracts to identify a date for which we switch from the front to second month contract for the spot market (second to third month contract for the futures) so as to piece together a daily time-series data set. For the NYMEX WTI futures, open interest in the front contract diverges, and the second month contract jumps, usually with 8 trading days to maturity for the front contract. We designate 8 trading days to maturity as the switch date to construct our sample⁵.

2.5 Basic details on the BBD (2016) EPU index

2.6 A market state variable BMC_t

Fama and French (1987) outline two popular theories for commodity futures pricing: The theory of storage proposed by Kaldor (1939), Working (1948), Brennan (1958) and Telser (1958), and the expectations theory of Cootner (1960), Dusak (1973) and Breeden (1980). Pindyck (1994) documents the association among low inventory level, high marginal convenience yield and a negative term slope. As inventory builds, convenience yield falls, and the term slope becomes positive. Considine et al. (2022) provide a comprehensive review of how studies of both theories relate inventory, convenience yield and market states. However, they

⁵After 8 days, the second and third contracts correspondingly become front and 2nd month contract, and the procedure is repeated.

use the sign of the basis to determine whether the crude oil market is in backwardation or contango.

Our conceptual framework defines backwardation and contango as $F_t < E_t(S_T)$ and $F_t > E_t(S_T)$ respectively. This allows a mean-variance hedger to ascertain if $E_t(r_{f,t+1})$ is greater or less than zero. We outline the term slope $TS_t = F_{t,T} - S_t$ under both theories.

$$\text{Expectation theory: } F_{t,T} - S_t = (F_{t,T} - E_t[S_T]) + (E_t[S_T] - S_t)$$

$$\text{Storage theory: } F_{t,T} - S_t = S_t R(t, T) + U(t, T) - Y(t, T)$$

Under expectations theory, TS_t consists of an expected risk premium $F_{t,T} - E_t[S_T]$ and a forecast of the future expected spot price $E_t[S_T] - S_t$. Since spot and futures prices must converge at contract maturity i.e. $E_t[S_T] = E_t[F_T]$, the sign of $(F_{t,T} - E_t[F_T])$ indicates to a mean-variance hedger whether $E_t(r_{f,t})$ is greater or less than zero. However, TS_t also depends on how $E_t[S_T]$ is expected to move from S_t . Since $E_t[S_T]$ is not observable at time t , we cannot readily infer the sign of $(F_{t,T} - E_t[S_T])$ from TS_t .

Under storage theory, the spot and futures pricing link is governed by arbitragers, who balance the physical carry costs of interest $S_t R(t, T)$ and storage $U(t, T)$ against the convenience benefit $Y(t, T)$ of physical ownership. We use TS_t to indicate the net benefit of ownership i.e. when $TS_t < 0$, the net benefit of ownership is positive $Y(t, T) > S_t R(t, T) + U(t, T)$, and vice versa. We cross reference TS_t with EIA inventory data to confirm whether there is indeed a supply shortage in the spot oil market. Specifically, the U.S. Energy Information Administration (EIA) releases crude oil stockpile data every Wednesday morning. We demean the stock series with a 3-year moving average to construct a cumulative change in weekly crude oil inventory INV_t in the U.S.

In sum, if price data indicates a net benefit in physical ownership $TS_t < 0$, and inventory data indicates a mean-adjusted decline in stockpile $INV_t < 0$, we interpret this as a strong signal of supply shortage in the crude oil spot market. This pushes S_t too high relative to F_t , and $E_t(S_T)$ is likely to fall substantially $F_{t,T} > E_t[S_T] = E_t[F_T]$ i.e. Contango. But if $TS_t < 0$ and $INV_t > 0$, this is a mixed signal on how $E_t[S_T]$ would move. When this happens, we set $F_{t,T} = E_t[S_T] = E_t[F_T]$ i.e. martingale. Conversely, if $TS_t > 0$ and $INV_t > 0$, this is a strong indication that there is currently an excess supply of crude oil, which depresses S_t substantially below F_t . Hence $E_t(S_T)$ is likely to rise substantially, such that $F_{t,T} < E_t[S_T] = E_t[F_T]$ i.e. Backwardation. If $TS_t > 0$ but $INV_t < 0$, this is also a mixed signal on how $E_t[S_T]$ would move, and so we set $F_{t,T} = E_t[S_T] = E_t[F_T]$ i.e. martingale.

Alternative is for $TS_t < 0$ and $INV_t > 0$ to indicate backwardation rather than martingale; For a negative term slope, we assume a moderate fall in $E_t(S_T)$ between S_t and F_t , such that $F_{t,T} < E_t[S_T]$ i.e. backwardation. Similarly, for $TS_t > 0$ and $INV_t < 0$, we assume a moderate rise in $E_t(S_T)$ between S_t and F_t , such that $F_{t,T} > E_t[S_T]$ i.e. contango.

INSERT FIGURE Y: Weekly term spread and cumulative change in stock level

Figure Y plots TS_T and INV_T over the full sample period, which shows comovement in the two series over time. Below, we follow the Fama and French (1987) test on whether i) realized premium is predicable i.e. time-varying expected risk premium; ii) the term slope can predict spot return at time T . Our main interest is on the estimated coefficients β_1, γ_1 i.e. whether TS_t or INV_t can indicate if the WTI futures market is in backwardation or contango.

$$\begin{aligned}(F_{t,T} - S_T) &= \alpha_1 + \beta_1 TS_{t,T} + \gamma_1 INV_t + \epsilon_{t1} \\ (S_T - S_t) &= \alpha_2 + \beta_2 TS_{t,T} + \gamma_2 INV_t + \epsilon_{t2}\end{aligned}\tag{4}$$

We use a combination of term slope and stockpile data to construct a market state variable BMC_t for the WTI crude oil futures market.

From the cost-of-carry pricing framework, term slope proxies the net benefit of futures ownership in lieu of physical ownership $S_t = F_t e^{(r+u-y)(T-t)}$, where r is the risk-free rate, u is unit storage cost, y is the implied convenience yield associated with spot ownership, and $(T-t)$ is the futures contract term to maturity. We proxy S_t with the current month futures price, and compare against the 1-year contract i.e. an 11-months term slope.

When there is excess demand for physical oil, this pushes S_t above F_t , which corresponds to a high $y > (r + u)$ i.e. negative term slope. The cumulative change in inventory is also likely to decline. If $S_t > F_{t,T}$, then $E_t(r_{f,T}) > 0$; As contract maturity $T - t$ declines, it gradually moves up a negative term slope. Furthermore, given the net benefit of spot ownership $y > (r + u)$, $E_t(r_{f,T}) > 0$ is required to attract long investors from spot to futures ownership. Short hedger are also willing to pay a ‘fee’ to defer giving up the net benefit of physical ownership to a later date. These market conditions are consistent with a backwardation state.

In contrast, when the physical oil market experiences excess supply, S_t falls below F_t , and reduced net benefit of spot ownership $y < (r + u)$ i.e. positive term slope. The cumulative change in stockpile is also likely to rise. If $S_t < F_{t,T}$, then $E_t(r_{f,T}) < 0$ given the contract is now riding down a positive term slope. Given the net benefit of futures ownership $y < (r + u)$, long hedgers are required to pay a ‘fee’ in $E_t(r_{f,T}) < 0$ to manage risk using futures, rather than to incur the cost of carrying inventory. These market conditions are consistent with a contango state.

we construct a market state variable BMC_t using the interaction between Term slope and stockpile inventory. We demean the time-series of weekly stockpile data to compute a cumulative abnormal stockpile CAS_t . Alongside a weekly term slope $TS_t = F_{t,T} - S_t$, we generate BMC_t as follows. When $TS_t < 0$ and $CAS_t < 0$, we interpret this as excess demand in the WTI crude oil spot market, which we classify as a backwardation state $BMC_t = B$; When $TS_t > 0$ and $CAS_t > 0$, we interpret this as excess supply in the spot market, which

we classify as a contango state $BMC_t = C$. And when the two variables have opposite signs, we classify this as a martingale state $BMC_t = M$. When

3 Empirical analysis and findings

Our empirical analysis has two main segments. First, we want to ascertain that EPU_t , as an economic state variable, has a significant impact on key variables relating to hedging WTI crude oil price risk, including spot-futures return, volatility, covariance and in-sample hedge ratios. In particular, we need to confirm that the statistical properties of in-sample hedge ratios are significantly different, when the U.S. economy goes through periods of high and low EPU states. The in-sample evidence motivates the application of EPU in state dependent hedging of WTI crude oil price risk. The statistical properties could also vary across hedge ratios estimated from different econometric models, which affects the specific form of state-dependent hedging to implement. As part of the in-sample analysis, we need to identify EPU thresholds associated with different EPU states.

Second, we perform out-of-sample evaluation of state-dependent hedging against both static and dynamic strategies. The evaluation approach is designed to highlight the potential benefit of incorporating EPU into hedging WTI crude oil price risk, regardless of the actual hedging configuration e.g. estimation window, econometric model etc.

3.1 EPU and in-sample hedge ratios

We utilize several econometric models to generate hedge ratios, including OLS, error-correction (EC), GARCH and stochastic volatility (SV). Denote the various models' hedge ratio as $\{\beta_{ols}, \beta_{ec}, \beta_{dcc}, \beta_{sv}\}$. The intention is not to conduct a comprehensive model horse-race, but to provide a coverage of commonly used models to estimate hedge ratio. We choose EC, GARCH and SV models because they are three classes of models that represent breakthroughs in the literature. For the model outline, we explain why we choose the Engle (2002) DCC-GARCH model, and Heston (1993) stochastic volatility model. More importantly, our paper aims to demonstrate the economic significance of incorporating EPU into a crude oil hedging strategy. For the out-of-sample evaluation, our methodology highlights potential benefits of state-dependent hedging, regardless of the econometric model that is used to generate hedge ratios.

For each model, we generate in-sample hedge ratios from 1997-2010. Using 75/25 break-points on the EPU_t distribution, we form hedge ratio sub-sample corresponding to high, medium and low EPU states. We perform statistical tests on the hedge ratio properties among EPU sub-samples. These results guide us to implement state-dependent hedging out-of-sample, and how we evaluate it against the static and dynamic strategies as benchmarks. The outline for the OLS hedge ratio in the next section generally applies to the EC, GARCH

and SV models as well. Where there are differences, we highlight them in the relevant section.

3.1.1 OLS hedge ratio β_{ols}

In equation (5), OLS is the most commonly used method to estimate hedge ratio β_{ols} . Based on a 6-years estimation window 1990-1996, we generate 14 years worth of in-sample daily hedge ratios $\beta_{ols,t}$ from 1997 to 2010. This gives around 12 years worth of data 2011-2022 to evaluate out-of-sample hedging performance. For preliminary testing, we apply ad-hoc 75/25 breakpoints on the EPU_t distribution for 1997-2010 to partition $\beta_{ols,t}$ into sub-samples associated with high $\beta_{ols,t}^h$, medium $\beta_{ols,t}^m$ and low $\beta_{ols,t}^l$ EPU states.

$$r_{s,t} = \alpha + \beta_{ols} r_{f,t} + \varepsilon_{s,t} \quad (5)$$

Figure 1A plots the distribution of $\beta_{ols,t}$, as well as $\{\beta_{ols,t}^h, \beta_{ols,t}^m, \beta_{ols,t}^l\}$. The range of hedge ratio values are comparable across EPU states, but β_{ols}^h and β_{ols}^l have visibly different distributions; β_{ols}^l has a generally more even distribution, while β_{ols}^h has more observations near the ends of the value range, and the least observations near the mid-point value around 1.07. The patterns in Figure 1A are more or less similar for estimation windows of 4 years (N=1000) in Figure 1B and 8 years (N=2000) in Figure 1C. Hence we focus our analysis on N=1500.

INSERT FIGURE 1 HERE

In Table 1, we conduct various tests on the statistical properties of in-sample hedge ratios. We have two main considerations: i) Does EPU affect the mean or variance of β_{ols} ? ii) Does β_{ols} behave differently across low, medium and high levels of EPU? Or, does it differ only between high and non-high EPU states? The specific nature of the relation between hedge ratios and EPU states may also differ across econometric models. Table 1A presents the descriptive statistics of $\{\beta_{ols}, \beta_{ols}^h, \beta_{ols}^m, \beta_{ols}^l\}$, as well as other key variables, including β_{ols} , spot volatility σ_s , futures volatility σ_f and spot-futures correlation ρ . To be consistent with β_{ols} , we estimate σ_s, σ_f, ρ based on a 6-years daily rolling window, and apply 75/25 EPU breakpoints to form EPU sub-samples for testing.

INSERT TABLE 1 HERE

In Table 1B, we run F-test and ANOVA analysis on the mean and variance of OLS hedge ratios among the three EPU sub-samples. This is to ascertain whether EPU_t affects the conditional mean or volatility of $\beta_{ols,t}$. The results confirm that β_{ols} , volatility and correlation are significantly different between the high and low EPU states. The significant difference holds for 4 years (N=1000) and 8 years (N=2000) estimation windows. **Add more discussion for actual Table 1**

3.1.2 Error-correction hedge ratio β_{ec}

Ghosh (1993) is one of the earlier papers to incorporate an error-correction (EC) mechanism between S_t and F_t into hedge ratio estimation. Spot and futures prices are I(1) non-stationary series, but share a cointegrating pricing link, such that an EC variable $u_t = S_t - a - bF_t$ is stationary.

$$\begin{aligned}
S_t &= a + bF_t + u_t \\
r_{s,t} &= \phi_{s0} + \beta_{ec}r_{f,t} + \lambda_s u_{t-1} + \sum_{k=1}^K (\phi_{ssk}r_{s,t-k} + \phi_{sfk}r_{f,t-k}) + \varepsilon_{s,t} \\
r_{f,t} &= \phi_{f0} + \beta_f r_{s,t} + \lambda_f u_{t-1} + \sum_{k=1}^K (\phi_{fsk}r_{s,t-k} + \phi_{ffk}r_{f,t-k}) + \varepsilon_{f,t}
\end{aligned} \tag{6}$$

From the Engel and Granger (1987) representation theorem, we apply a two-step estimation procedure on the bivariate VECM in equation (6), with coefficient restrictions $\beta_{ec} = \beta_f = 0$. The first step is a price-level regression to extract the residual u_t , which enters as a lag-1 EC variable into the second step VECM estimation on $r_{s,t}$ and $r_{f,t}$. The lag specification K is determined by Akaike information criterion (AIC) to ensure that the residuals $\varepsilon_{s,t}, \varepsilon_{f,t}$ are white noise. Ghosh (1993) estimates only the $r_{s,t}$ equation using OLS to obtain the error-correction hedge ratio β_{ec} . In contrast, Kroner and Sultan (1993) fit a bivariate GARCH on $\varepsilon_{s,t}$ and $\varepsilon_{f,t}$, which are derived from (6) as the mean equations with $\beta_{ec} = \beta_f = 0$. They estimate hedge ratios from the GARCH conditional futures variance and spot-futures covariance.

Since we already have DCC-GARCH hedge ratios, we see marginal value in examining VEC-GARCH hedge ratios proposed by Kroner and Sultan (1993). Instead, we estimate two EC-adjusted hedge ratios, conduct in-sample evaluation, and apply the better version for the main analysis. The first is to follow Ghosh (1993) and estimate β_{ec} from OLS rolling-regression on $r_{s,t}$. The second is to apply a FIML rolling estimation on the bivariate VECM in equation (6), with $\beta_{ec} = \beta_f = 0$. For each estimation round, we compute a hedge ratio using the variance-covariance matrix for $\varepsilon_{s,t}, \varepsilon_{f,t}$.

3.1.3 Stochastic volatility (SV) hedge ratio β_{sv}

We outline the Heston (1993) continuous-time stochastic volatility (SV) model in equation (7), where V_t is the volatility process for movements in asset price S_t , and μ is the expected return. Brownian increments for the price ∂W_t^S and volatility process ∂W_t^V are correlated with $E(\partial W_t^S \partial W_t^V) = \rho dt$. The Black (1976) leverage effect implies $\rho < 0$, as price falls induce

more volatility compared to price rises. ∂V_t captures serial correlation in volatility, with the mean-level α_0 and speed of mean-reversion α_1 . Lastly, η can be viewed as a ‘volatility-of-volatility’ parameter, which affects the thickness in the tails of the return distribution.

$$\begin{aligned}\frac{\partial S_t}{S_t} &= (\mu + cV_t)\partial t + \sqrt{V_t}\partial W_t^S \\ \partial V_t &= (\alpha_0 - \alpha_1 V_t)\partial t + \eta\sqrt{V_t}\partial W_t^V \\ \beta_{sv,t} &= \rho_{sf,t}\left(\frac{\sqrt{V_{s,t+1}}}{\sqrt{V_{f,t+1}}}\right)\end{aligned}\tag{7}$$

Denote the SV hedge ratio as $\beta_{sv,t}$, where $\phi_{sf,t}$ is the Pearson conditional correlation between spot and futures return from rolling window estimation. Other well-known SV models in the literature include the Bates (1996) SV model with price jumps, and the Duffie et al. (2000) SV model with correlated jumps. Our aim is not to provide a comprehensive model horse-race, but to demonstrate the economic significance of incorporating EPU into a crude oil hedging strategy. Our out-of-sample evaluation approach highlights the potential benefits of state-dependent hedging, regardless of the specific econometric model that is used to generate hedge ratios. As such, the original SV model of Heston (1993) suffices as a benchmark.

3.1.4 DCC-GARCH hedge ratio β_{dcc}

We repeat the in-sample analysis for hedge ratios generated from the Engle (2002) dynamic conditional correlation (DCC) GARCH model. The DCC estimator generalizes the Bollerslev (1990) constant conditional correlation (CCC) estimator to allow the correlation matrix to vary over time based on GARCH estimated parameters. From the huge family of GARCH models, we choose DCC-GARCH for two reasons. First, we want to generate hedge ratios from a multivariate GARCH model that is not complex to estimate, but yet has been shown to parsimoniously fit the data for many asset markets. Second, the DCC estimator emphasises a time-varying correlation structure. This blends well with the concept of state-dependent hedging, which stipulates a changing covariance structure associated with a relevant economic state or market condition. Equation (8) outlines how we estimate hedge ratio from a bivariate

DCC-GARCH.

$$\begin{aligned}
r_{s,t} &= \phi_{s0} + \sum_{k=1} \phi_{sk} r_{s,t-k} + \varepsilon_{s,t} \\
r_{f,t} &= \phi_{f0} + \sum_{k=1} \phi_{fk} r_{f,t-k} + \varepsilon_{f,t} \\
\sigma_{s,t}^2 &= a_{s0} + a_{s1} \varepsilon_{st-1}^2 + b_s \sigma_{s,t-1}^2 \\
\sigma_{f,t}^2 &= a_{f0} + a_{f1} \varepsilon_{ft-1}^2 + b_f \sigma_{f,t-1}^2 \\
\sigma_{sf,t} &= a_{sf0} + a_{sf1} \varepsilon_{st-1} \varepsilon_{ft-1} + b_{sf} \sigma_{sf,t-1} \\
\beta_{dcc,t} &= \frac{\sigma_{sf,t+1}}{\sigma_{f,t+1}^2}
\end{aligned} \tag{8}$$

Figure 2 plots the distribution of DCC-GARCH hedge ratios β_{dcc} , as well as sub-samples associated with the three EPU states. In Table 2A, we present descriptive statistics for $\{\beta_{dcc}, bd^h, bd^m, bd^l\}$. Panel B outlines test statistics on whether the mean and variance between bd^h, bd^l are significantly different. **Discuss on actual F2 and T2.**

INSERT FIGURE 2 HERE

INSERT TABLE 2 HERE

The above analysis confirms that model-specific hedge ratios exhibit significantly different statistical properties across EPU states.

3.2 Identify pre-specified EPU thresholds

To implement state-dependent hedging, a pre-specify EPU threshold is required to guide the switch between static and dynamic strategies, as the U.S. economy transits through different EPU states during the 2011-2022 out-of-sample period. The actual hedging configuration could possibly vary across hedge ratio models along two dimensions: i) The number of pre-specified EPU thresholds; ii) Whether EPU affects the mean or volatility of hedge ratios, or both? These two dimensions determine the number of EPU thresholds to apply, and the exact form of state-dependent hedging to implement, when a threshold is crossed. They relate back to the in-sample analysis on how different EPU states affect the statistical properties of in-sample hedge ratios from various econometrics models.

Take β_{ols} for example; Assume only the high EPU threshold EPU^h is relevant, and it affects the level of β_{ols} , but not its volatility i.e. $\mu_{\beta_{ols}^h} \neq \mu_{\beta_{ols} \setminus \beta_{ols}^h}$, but $\sigma_{\beta_{ols}^h} = \sigma_{\beta_{ols} \setminus \beta_{ols}^h}$. In this scenario, state-dependent hedging aligns closer with a static strategy; The hedger re-balances

only when the U.S. economy enters into, or exists from a state of high EPU i.e. when EPU_t crosses EPU^h . If two thresholds are relevant i.e. $\mu_{\beta_{ols}^h} \neq \mu_{\beta_{ols}^m} \neq \mu_{\beta_{ols}^l}$, the hedger re-balances whenever EPU_t crosses either EPU^h or EPU^l , but would otherwise maintain a static hedge.

Now consider if EPU affects OLS hedge ratio volatility $\sigma_{\beta_{ols}}$. Assume WTI crude oil hedge ratio becomes unstable when the U.S. economy enters into a state of high economic uncertainty i.e. $\sigma_{\beta_{ols}^h} > \sigma_{\beta_{ols} \setminus \beta_{ols}^h}$. In this scenario, the hedger maintains a static hedge when $EPU_t < EPU^h$, but switches to a dynamic hedge when $EPU_t \geq EPU^h$. This is due to a higher $\sigma_{\beta_{ols}^h}$ increasing the hedger's re-balancing option value. And when EPU_t falls back below EPU^h , the conditional hedge ratio stabilizes. This reduces the value of the re-balancing option, so the hedger restores a static hedge. Similarly, if $\sigma_{\beta_{ols}^h} > \sigma_{\beta_{ols}^m} > \sigma_{\beta_{ols}^l}$, the re-balancing option value increases with EPU_t . The hedger maintains a static hedge during the low EPU state, and switches to weekly or fortnightly re-balancing during the medium EPU state. When the economy enters a high EPU state, the hedger re-balances in response to changes in the optimal number of futures contracts over time.

The current discussion shows that the specific hedging configuration differs across scenarios in terms of re-balancing frequency. This relates back to the economic value of state-dependent hedging. If EPU does not significantly affect hedge ratio mean or volatility, it cannot guide a hedger's re-balancing decision. If the impact of EPU on hedge ratio is felt over a wider range of EPU_t levels i.e. from 0 to 1 to 2 or more thresholds, the expected re-balancing frequency increases. When rises in EPU also increases hedge ratio volatility i.e. from low to medium to high EPU state, this further increases the re-balancing frequency, from static to weekly or fortnightly, to as required by changes in the optimal futures position. As a conditional hedge ratio becomes more volatile, it increases the hedger's re-balancing option value. Given the same proportional option premium (transaction cost in terms of brokerage fee, margin requirement and capital commitment to meet margin calls etc), the greater (lower) the volatility in hedge ratio, the greater (lower) the likelihood for a hedger to take up the option to re-balance the hedging position.

The earlier in-sample analysis on hedge ratios from various models provide results to ascertain the number of EPU threshold(s), and whether EPU affects the mean or volatility of hedge ratios. We outline two approaches to identify EPU thresholds.

3.2.1 EPU breakpoints implied from in-sample hedge ratio

Using the in-sample hedge ratio distribution from 1997 to 2010, we numerically solve for a threshold value that maximizes the difference in the mean μ (or volatility σ) between EPU sub-samples. Depending on the in-sample analysis for each model, our numerical search criterion focuses on either μ or σ , and to seek 1, 2 or more EPU thresholds. We provide a detailed outline for β_{ols} , which generally applies to β_{ec} , β_{dcc} and β_{sv} as well.

The in-sample analysis on β_{ols} reveals that it is significantly more volatile during the

high EPU state, compared to normal or low EPU states. Accordingly, we search for EPU^h that maximizes the difference between the volatility of the β_{ols}^h sub-sample ($\sigma_{\beta_{ols}^h}$), and that of the remaining OLS hedge ratios $\sigma_{\beta_{ols} \setminus \beta_{ols}^h}$. Since the in-sample analysis is based on 75/25 breakpoints, we use the 75th percentile as the initial value for the numerical search. To ensure that a hedge ratio sub-sample contains a minimal number of observations, we impose constraints on the numerical solutions for EPU^h and EPU^l by setting 90/10 percentiles as the strict upper and lower bounds respectively.

If instead the in-sample analysis had indicated that the mean of β_{ols} is significantly different across 3 EPU states i.e. $\mu_{\beta_{ols}^h} \neq \mu_{\beta_{ols}^m} \neq \mu_{\beta_{ols}^l}$, we would numerically search for a pair of thresholds EPU^h, EPU^l which maximize the sum of magnitude difference between $\mu_{\beta_{ols}^h}, \mu_{\beta_{ols}^m}$, and between $\mu_{\beta_{ols}^m}, \mu_{\beta_{ols}^l}$. The 75/25 breakpoints are used as starting seeds, and the numerical solutions are bounded by the 90/10 percentiles.

3.2.2 EPU statistical bounds

We outline a second approach in equation (9), which derives EPU^h, EPU^l as statistical bounds from the mean μ_{epu} and standard deviation σ_{epu} of EPU_t over the in-sample period 1990-2010. The scaling parameters λ^h, λ^l serve two purposes: i) As a robustness check; We generate baseline results from $\lambda^h = \lambda^l = 1$, and check for consistency across $\lambda = \{0.8, 0.9, 1.1, 1.2\}$; ii) To allow for asymmetric thresholds between high and low EPU states e.g. a weaker threshold $\lambda^h = 0.8$ for high EPU state, and a stronger threshold $\lambda^l = 1.2$ for low EPU state.

$$\begin{aligned} EPU^h &= \mu_{epu} + \lambda^h \sigma_{epu} \\ EPU^l &= \mu_{epu} - \lambda^l \sigma_{epu} \end{aligned} \tag{9}$$

The two approaches are comparable in that EPU thresholds are determined based on the 1990-2010 in-sample period, and applied to state-dependent hedging over the 2011-2022 out-of-sample period. They can be used to identify either 1 or 2 thresholds, whether symmetric or asymmetry. Lastly, both methods allow for conditional thresholds over time. For example, after the two year of hedging, we can forward shift the in-sample period from 1990-2010 to 1992-2012, compute $(\mu_{epu}, \sigma_{epu})$ to update EPU^h, EPU^l for hedging in 2013, and so forth.

4 Out-of-sample hedging performance

The in-sample analysis confirms that hedge ratios from all four econometric models exhibit significantly different statistical properties across EPU states. These findings are based on an ad-hoc 75/25 break-point specification on EPU_t to form hedge ratio sub-samples. If we had

formally identified EPU breakpoints for the sub-sample testing, we would obtain stronger findings that EPU affects WTI hedge ratios.

In this section, we evaluate the out-of-sample performance of state-dependent hedging against both static and dynamic strategies as benchmarks. Our aim is to draw out the potential economic value from incorporating EPU into a hedger's re-balancing decision in managing WTI crude oil risk, *regardless* of the econometric model that is used to generate hedge ratios. With that in mind, we present two evaluation approaches in separate sub-sections. Before that, we outline the basic hedging implementation framework below.

4.1 The hedging framework

Following the literature, we implement hedging from the perspective of a commercial user who manages rising crude oil prices by taking long positions in NYMEX WTI futures contracts, each written on 1,000 (1K) barrels of crude oil. Assume the commercial hedger has a physical exposure of 10,000 (10K) barrels of WTI crude oil. The number of long futures contracts $N_{i,t}$ is given by equation (10), where $\beta_{i,t}$ is the conditional hedge ratio of hedging strategy i , $V_{s,t} = 10Ks_t$ is the value of the spot exposure at spot price s_t per barrel, and $v_{f,t} = 1Kf_t$ is the value covered by a contract at futures price f_t per barrel. Carry-cost parity pricing ensures that $\frac{s_t}{f_t}$ gravitates towards 1 over time. As such, re-balancing in response to changes in $N_{i,t}$ is mainly driven by fluctuations in $\beta_{i,t}$ over time.

$$N_{i,t} = \beta_{i,t} \left(\frac{V_{s,t}}{v_{f,t}} \right) = 10\beta_{i,t} \left(\frac{s_t}{f_t} \right) \quad (10)$$

$$\begin{aligned} V_{s,t} &= 10Ks_t \\ \Delta V_{s,t} &= 10Kr_{s,t} \\ V_{fit} &= v_{f,t}N_{i,t} = 1Kf_tN_{i,t} \\ \Delta V_{fit} &= 1K[f_tN_{i,t} - f_{t-1}N_{i,t-1}] \\ \Delta N_{i,t} &= N_{i,t} - N_{i,t-1} \\ \Delta V_{fit} &= 1K[f_tN_{i,t} - f_{t-1}N_{i,t} + f_{t-1}\Delta N_{i,t}] \\ &= 1K[r_{f,t}N_{i,t} + f_{t-1}\Delta N_{i,t}] \end{aligned}$$

$$r_{hit} = \Delta V_{s,t} - \Delta V_{fit} = 1K[10r_{s,t} - r_{f,t}N_{i,t} - f_{t-1}\Delta N_{i,t}] \quad (11)$$

To evaluate hedging performance, we compute and analyse the return of the commercial user's hedged portfolio over time r_{hit} in equation (11), where ΔV_{fit} is the change in value of the futures position based on hedging strategy i . In our paper, it is more appropriate to index i as hedge strategy rather than econometric model. This is because a state-dependent

hedge can be implemented based on e.g. $\beta_{dcc,t}$, and evaluate against a standard dynamic hedge using $\beta_{dcc,t}$. Each trading day, we mark-to-market the value of the hedger's spot $V_{s,t}$ and futures V_{fit} positions using daily settlement prices, and scale by the relevant quantity units. Using these two time-series, we compute the daily change in spot and futures value for the hedged portfolio's daily return $r_{hit} = \Delta V_{s,t} - \Delta V_{fit}$ for hedge strategy i . The analytical derivation of r_{hit} assumes that the spot quantity exposure is constant at 10K barrels; The futures quantity could vary over time, as indicated by $\Delta N_{i,t}$. When there is no rebalancing on day t , $\Delta N_{i,t} = 0$, such that r_{hit} is affected only by spot $r_{s,t}$ and futures return $r_{f,t}$. When contracts are either opened or closed off, we deduct from r_{hit} 0.3% per contract trade as transaction cost.

Using the outlined hedging framework, we can compute the hedged portfolio daily returns for any hedge strategy i , over the 2011-2022 out-of-sample period. This allows us to evaluate state-dependent hedging performance against relevant benchmark hedge strategies. In terms of performance measure, we consider both the Ederington (1979) risk-reduction and Howard and D'Antonio (1984) Sharpe Ratio (SR) enhancement, against that of relevant benchmark hedge strategies. Now, we outline two benchmarking approaches that are designed to evaluate the economic value of incorporating EPU to hedge against WTI crude oil risk.

4.2 Switch between static and dynamic hedge

This approach evaluates a hedge ratio against itself, in order to remove or substantially reduce the influence of an econometric model on state-dependent hedging performance. We elaborate based on DCC-GARCH hedge ratio β_{dcc} , and assume only EPU^h is relevant. The approach similarly applies to $\{\beta_{ols}, \beta_{ec}, \beta_{sv}\}$, and it can accommodate 2 or more EPU thresholds. We estimate a DCC-GARCH from 2004-2010 to compute an initial hedge ratio and set up a benchmark static hedge for the 2011-2022 out-of-sample period. To construct a dynamic hedge benchmark, we roll the DCC-GARCH estimation to generate a daily time-series of $\beta_{dcc,t}$ from 2011 to 2022, and re-balance in response to changes in the optimal number of futures contracts over time.

The exact form of state-dependent hedging depends on the statistical properties of in-sample $\beta_{dcc,t}$ under different EPU states. We outline below the possible scenarios, and the corresponding form of state-dependent hedging. To note, these scenarios are based on analysing the distribution of in-sample hedge ratios from 1997 to 2010. As hedging is implemented gradually over the 12 years out-of-sample period, we can update test results on the statistical properties of in-sample hedge ratios (say) every 2 years from 1997-2010 to 1999-2012, update the applicable scenario for each model's hedge ratio, and apply the corresponding state-dependent hedge for the next two years 2013-2014, and so forth.

1. Insignificant difference in mean and variance. $\mu_{hi,t}^h = \mu_{hi,t} \setminus \mu_{hi,t}^h$; $\sigma_{hi,t}^h = \sigma_{hi,t} \setminus \sigma_{hi,t}^h$: EPU is not a relevant state variable for $\beta_{dcc,t}$.

2. Significant difference in mean $\mu_{h_{i,t}}^h \neq \mu_{h_{i,t}} \setminus \mu_{h_{i,t}}^h$; Insignificant difference in variance $\sigma_{h_{i,t}}^h = \sigma_{h_{i,t}} \setminus \sigma_{h_{i,t}}^h$: Re-balance only when EPU_t crosses EPU^h .
3. Insignificant difference in mean $\mu_{h_{i,t}}^h = \mu_{h_{i,t}} \setminus \mu_{h_{i,t}}^h$; Significantly larger variance $\sigma_{h_{i,t}}^h > \sigma_{h_{i,t}} \setminus \sigma_{h_{i,t}}^h$:
 - Perform variance decomposition on the higher volatility sub-sample $h_{i,t}^h$.
 - Weights to conditional covariance $\sigma_{sf,t}$ and futures conditional variance $\sigma_{f,t}^2$.
 - If volatility mainly from $\sigma_{sf,t}$, switch to dynamic hedge.
 - If volatility mainly from $\sigma_{f,t}^2$, maintain static hedge.
4. Significant difference in mean and variance $\mu_{h_{i,t}}^h \neq \mu_{h_{i,t}} \setminus \mu_{h_{i,t}}^h$; $\sigma_{h_{i,t}}^h > \sigma_{h_{i,t}} \setminus \sigma_{h_{i,t}}^h$: Static when $EPU_t < EPU^h$; Switch to dynamic when $EPU_t > EPU^h$.

The first scenario is obvious. If EPU affects neither $\mu_{\beta_{dcc,t}}$ nor $\sigma_{\beta_{dcc,t}}$, it is not relevant for state-dependent hedge implementation based on β_{dcc} . In the second scenario, the level of β_{dcc} is significantly different during a state of high EPU. Accordingly, a hedger re-balances (if needed) whenever EPU_t crosses EPU^h . But given no significant increase in $\sigma_{\beta_{dcc,t}}$, there is limited justification to switch to a dynamic hedge strategy.

The third scenario considers similar levels, but more volatile hedge ratios during a state of high EPU. However, greater variability in $\beta_{dcc,t}$ does not necessarily imply better hedging performance. Lien (2010) derive an analytical expression for the return variance of a hedged portfolio, and show the conditions under which greater hedge ratio variability reduces the hedged portfolio risk. In equation (10), the risk-minimizing hedge ratio $\beta_{i,t}$ is the ratio between spot-futures return covariance $\sigma_{sf,t}$ and the futures variance $\sigma_{f,t}^2$. The hedge position is adjusted mainly in response to fluctuation in the return co-movement between the exposed asset and hedging instrument. Accordingly, if variability in $\beta_{dcc,t}$ is mainly attributed to time-varying spot-futures correlation, we regard this as informative, and switch to a dynamic strategy. But if variability in $\beta_{dcc,t}$ is mainly driven by $\sigma_{f,t}^2$, we regard this as noisy, and retain a static hedge. To note, the structure of $\beta_{i,t}$ is consistent with our interpretation; Assume a sudden rise in speculative trading the futures market, causing $\sigma_{f,t}^2$ to jump up, but without affecting $\sigma_{sf,t}$. This reduces the hedge ratio substantially, as the futures position would add, rather than reduce the hedged portfolio's volatility.

In the fourth and last scenario, if high EPU in the U.S economy affects both the mean and volatility of $\beta_{dcc,t}$, we regard this as a structural change. When $EPU_t \leq EPU^h$, a static hedge applies. And when $EPU_t > EPU^h$, the hedger switches to a dynamic strategy, following the same configuration as the benchmark dynamic hedge. Under this scenario, a state-dependent hedge is a hybrid of static and dynamic strategies that generates economic value by timing the EPU state-specific benefits associated with both hedging strategies; A static hedge is beneficial when $\beta_{dcc,t}$ is stable during low EPU states; A dynamic hedge is beneficial when $\beta_{dcc,t}$ becomes volatile during a high EPU state.

In the literature, some studies impose a re-balancing constraint on the dynamic strategy to curb transaction cost. Two common constraints are i) to apply a pre-specified re-balancing frequency e.g. weekly or fortnightly, and ii) to re-balance only when the optimal hedging position changes by 2 (or 3, 4 etc) contracts. These and other hedging configurations, are unlikely to affect our evaluation results; The hedging configuration for the dynamic hedge benchmark is also applied on the dynamic phase of state-dependent hedging. Given all 3 hedging strategies are based on the DCC-GARCH, we can attribute incremental hedging performance to EPU guiding the strategy switch.

4.2.1 OLS hedge ratio

4.2.2 EC hedge ratio

4.2.3 DCC-GARCH hedge ratio

4.2.4 SV hedge ratio

4.3 EPU-adjusted OLS vs. EC/DCC-GARCH/SV

Another evaluation approach that removes the model dimension is to formally derive EPU-modified OLS, EC, DCC-GARCH and SV models, generate ‘EPU-adjusted’ hedge ratios and evaluate against their unadjusted peers from the same model. While this is conceptually sound, it requires a non-trivial effort to formally modify all four econometric models, in particular for DCC-GARCH and SV models. Indeed, we could pursue such an avenue in a more technical paper that contribute to model development. In a concurrent paper, we incorporate the Zakoian (1994) TGARCH with EPU thresholds, into a DCC-GARCH model to generate hedge ratios with threshold effects in the DCC matrix that is delivered by fluctuations in EPU_t over time.

We consider a variant approach that is not onerous on model extension, but which can convincingly highlight the economic value of incorporating EPU into WTI crude oil hedging. We incorporate EPU-related variables into an OLS regression to generate an EPU-adjusted hedge ratio β_{ols}^e , which we directly benchmark against unadjusted hedge ratios from EC, DCC-GARCH and SV models. It is well-established in the literature that these models possess various statistical merits over an OLS regression. Despite its statistical shortcomings, if β_{ols}^e can hold its own hedging performance against superior but unadjusted benchmark hedge ratios, this constitutes strong evidence on the economic value of state-dependent hedging. We outline two modified OLS regressions to generate EPU-adjusted hedge ratios.

$$r_{s,t} = \alpha + [\beta_{ols}^e + \beta'_{ols} EPU_t] r_{f,t} + \epsilon_t \quad (12)$$

In equation (12), we add an interacting variable between EPU_t and $r_{f,t}$ to the OLS

regression. Through β'_{ols} , EPU_t delivers a conditional effect on β^e_{ols} . As the estimation rolls over the 2011-2022 hedging period, there could be times when β'_{ols} becomes significant, or vice versa. It is also possible for a significant β'_{ols} to change sign, depending on whether EPU_t fluctuates towards a high or low EPU state during the estimation window. Using the conditional estimates from equation (12), we implement state-dependent hedging as follows: If $\beta'_{ols,t}$ is insignificant, the OLS hedge is $\beta^e_{ols,t}$; When $\beta'_{ols,t}$ becomes significant, the OLS hedge is $\beta^e_{ols,t} + \beta'_{ols,t}$. Re-balancing occurs whenever the EPU-adjusted OLS hedge ratio changes in the optimal number of futures contracts to hold.

$$\begin{aligned}
r_{s,t} &= \alpha + (\beta^{ed}_{ols} + \beta^h_{ols}Dum_t^h + \beta^l_{ols}Dum_t^l)r_{f,t} + \epsilon_t \\
Dum_t^h &= \begin{cases} 1, & \text{if } EPU_t > EPU^h \\ 0, & \text{otherwise,} \end{cases} \\
Dum_t^l &= \begin{cases} 1, & \text{if } EPU_t < EPU^l \\ 0, & \text{otherwise.} \end{cases} \tag{13}
\end{aligned}$$

We outline a second modified OLS regression in equation (13), which utilizes threshold dummies Dum_t^h and Dum_t^l to generate EPU-adjusted hedge ratios. In an earlier section, we outline methods to identify the EPU thresholds EPU^h, EPU^l from in-sample analysis. To further simplify the extension, we could impose ad-hoc 75/25 percentiles from EPU_t as thresholds. The re-balancing decision is guided by the dummy coefficients β^h_{ols} and β^l_{ols} . Consider an unconditional estimation of equation (13) for the 1990-2010 in-sample period. Assuming both Dum_t^h, Dum_t^l are significant, out-of-sample re-balancing is guided by the following 3 hedge ratios, depending on how EPU_t fluctuates against its thresholds⁶.

1. $\beta^{ed}_{ols} + \beta^h_{ols}$, when $EPU_t \geq EPU^h$
2. β^{ed}_{ols} , when $EPU^l < EPU_t < EPU^h$
3. $\beta^{ed}_{ols} + \beta^l_{ols}$, when $EPU_t \leq EPU^l$

The EC/GARCH/SV hedge ratios are derived from conditional estimates. For consistency, we conduct a rolling estimation of equation (13) to generate EPU-adjusted conditional hedge ratios. Indeed, we could also update the thresholds EPU^h, EPU^l (say) annually, as the hedging progresses over the 2011-2022 out-of-sample period.

⁶If Dum_t^l (Dum_t^h) is insignificant, then the EPU-adjusted OLS hedge ratio is simply β^{ed}_{ols} when $EPU_t \leq EPU^l$ ($\geq EPU^h$).

4.3.1 Out-of-sample hedging effectiveness

We evaluate the two EPU-adjusted OLS hedge ratios β_{ols}^e and β_{ols}^{ed} against all four unadjusted hedge ratios $\{\beta_{ols}, \beta_{ec}, \beta_{dcc}, \beta_{sv}\}$ as benchmarks.

4.4 Further analysis on out-of-sample hedging performance

Many hedging studies routinely perform robustness checks with different hedge performance measures, of which risk-reduction and Sharpe Ratio enhancement over the unhedged position are commonly used. While the relative rankings are usually consistent, which we also show in our paper, conceptually the approach is inappropriate. The performance measure has to align with the objective function from which the optimal hedge ratio is derived: Risk-reduction criterion for the Johnson (1976) and Edrington (1979) risk-minimizing hedge ratio; Sharpe Ratio criterion for the Kroner and Sultan (1983) mean-variance slope maximizing hedge ratio.

To address the importance of EPU to hedging, we only consider the risk-minimizing hedge ratio. Other than the standard risk-reduction measure of hedging effectiveness $HE = [1 - \frac{\sigma_{hedged}}{\sigma_{unhedged}}]$, we consider a second HE measure that is modified from Sanders and Manfredo (2004). They derive a HE measure based on the encompassing principle in the forecasting literature, and aligns it with a risk minimization objective. In equation (14), Sanders and Manfredo (2004) compare the residual hedging errors $\{\varepsilon_{a,t}, \varepsilon_{b,t}\}$ *i.e. basis risk* between two competing hedging contracts A and B, with returns $\{r_{a,t}, r_{b,t}\}$. The estimated coefficient λ_1 relates to the hedged portfolio weight for contract B. Intuitively, if contract A encompasses the hedging ability of contract B, $(\varepsilon_{a,t} - \varepsilon_{b,t})$ will exhibit low mean and volatility, such that λ_1 is insignificant, and contract B has zero weight in the hedged portfolio.

$$\begin{aligned} r_{s,t} &= \alpha_a + \beta_a r_{a,t} + \varepsilon_{a,t} \\ r_{s,t} &= \alpha_b + \beta_b r_{b,t} + \varepsilon_{b,t} \\ \varepsilon_{a,t} &= \lambda_0 + \lambda_1(\varepsilon_{a,t} - \varepsilon_{b,t}) + v_t \end{aligned} \tag{14}$$

The Sanders and Manfredo (2004) measure is useful for evaluating competing hedging contracts, or to ascertain the complementary benefit from adding a second contract to a hedged portfolio. Conceptually, it is a suitable performance measure for our paper, given that state-dependent hedging encompasses elements of both static and dynamic strategies. We modify equation (14) to shift the emphasis from hedge contract $r_{a,t}$ vs. $r_{b,t}$, to hedge strategy $\beta_{i,t}$ vs. $\beta_{j,t}$ in equation (15). Our modified measure can evaluate out-of-sample hedging performance.

$$\begin{aligned}
r_{hit} &= \Delta V_{s,t} - \Delta V_{fit} \\
r_{hjt} &= \Delta V_{s,t} - \Delta V_{fjt} \\
r_{hit} &= \lambda_0 + \lambda_j(r_{hit} - r_{hjt}) + v_t
\end{aligned} \tag{15}$$

In equation (15), the strategy-specific hedged portfolio return r_{hit}, r_{hjt} represent hedge errors for the out-of-sample period. A hedge-error regression compares strategies i and j , where the residual v_t can be viewed as a net hedging error. If strategy i fully encompasses j , then λ_j is insignificant, and $v_t = r_{hit}$. Conversely, if strategy j fully encompasses i , then λ_j is significant and close to 1, such that $v_t = r_{hjt}$. Lastly, if neither i nor j fully encompasses the other, then $0 < \lambda_j < 1$, such that v_t is some weighted-average hedging error of both strategies.

$$\begin{aligned}
v_t &= r_{hit} - \lambda_0 - \lambda_j(r_{hit} - r_{hjt}) \\
\sigma_{v_t}^2 &= \sigma_{r_{hit}}^2 + \lambda_j^2 Var[r_{hit} - r_{hjt}] - 2\lambda_j Var[r_{hit}(r_{hit} - r_{hjt})] \\
\frac{\partial \sigma_{v_t}^2}{\partial \lambda_j^*} &= 2\lambda_j Var[r_{hit} - r_{hjt}] - 2Var[r_{hit}(r_{hit} - r_{hjt})] = 0 \\
\lambda_j^* &= \frac{\sigma_{r_{hit}}^2 - \rho_{r_{hit}, r_{hjt}} \sigma_{r_{hit}} \sigma_{r_{hjt}}}{\sigma_{r_{hit}}^2 + \sigma_{r_{hjt}}^2 - 2\rho_{r_{hit}, r_{hjt}} \sigma_{r_{hit}} \sigma_{r_{hjt}}} \begin{cases} \geq 0, & \text{if } \rho_{r_{hit}, r_{hjt}} \leq \frac{\sigma_{r_{hit}}}{\sigma_{r_{hjt}}} \\ < 0, & \text{if } \rho_{r_{hit}, r_{hjt}} > \frac{\sigma_{r_{hit}}}{\sigma_{r_{hjt}}} \end{cases} \tag{16}
\end{aligned}$$

In equation (16), we derive the optimal λ_j^* which minimizes the variance of the net hedging error $\sigma_{v_t}^2$. The functional form reveals a trade-off between hedge strategies i, j , which we conceptualize as follows. The measure λ_j^* benchmarks strategy j against strategy i in terms a hedge-error variance-ratio $\frac{\sigma_{r_{hit}}}{\sigma_{r_{hjt}}}$ and a hedge-error correlation $\rho_{r_{hit}, r_{hjt}}$. For strategy j to ‘add value’ to strategy i ($\lambda_j > 0$), either $\frac{\sigma_{r_{hit}}}{\sigma_{r_{hjt}}}$ has to be sufficiently large, or $\rho_{r_{hit}, r_{hjt}}$ has to be sufficiently small. For example, if the hedge errors are highly correlated $\rho_{r_{hit}, r_{hjt}} \approx 1$, then strategy j would be beneficial over strategy i if it exhibits a lower hedge error volatility i.e. $\frac{\sigma_{r_{hit}}}{\sigma_{r_{hjt}}} > 1$. The lower the hedge-error correlation, and the higher the hedge-error variance-ratio, the more beneficial is strategy j to a hedger compared to strategy i , which corresponds to a larger λ_j . Consider two extreme cases; When $\sigma_{r_{hit}, r_{hjt}} = \sigma_{r_{hit}}^2$, $\lambda_j = 0$; If the two strategies’ hedge-error covariance is equivalent to the hedge-error variance of strategy i , strategy j exhibits the same hedge-error volatility as i , hence offers no added benefit. Conversely, if $\sigma_{r_{hit}, r_{hjt}} = \sigma_{r_{hjt}}^2$, strategy i exhibits the same hedge-error volatility as j , such that it is replaced $\lambda_j = 1$.

Since λ_j is based on a pairwise comparison, we apply to the two hedging evaluation approaches as follows: The first approach uses EPU to guide the rebalancing decision, and compares hedge strategies based on the same econometric model. Take DCC-GARCH for example; We compute λ_j for a dynamic hedge $\beta_{dcc,t}$ as strategy j , using a static hedge β_{dcc} as benchmark i . Assume $\beta_{dcc,t}$ encompasses β_{dcc} . Next, we compute λ_j for a state-dependent DCC-GARCH hedge as strategy j , against the winning strategy $\beta_{dcc,t}$ as i . The same two-stage evaluation applies to state-dependent hedge strategies from other econometric models. The second approach compares each of the two EPU-adjusted OLS hedge ratios $j = \{\beta_{ols}^e, \beta_{ols}^{ed}\}$, against each of the four unadjusted benchmarks hedge ratios from OLS, EC, GARCH and SV models $i = \{\beta_{ols}, \beta_{ec}, \beta_{dcc}, \beta_{sv}\}$.

Structure of paper

There are two main parts: In-sample and Out-of-sample.

IN-SAMPLE: 1990-2010

1. Significant difference statistical properties for 75/25 EPU sub-samples: $\{r_t, \sigma_{st}, \sigma_{ft}, \sigma_{sf,t}, b_t, \sigma_{bt}\}$
2. Generate in-sample hedge ratios $\beta_{i,t}, i = OLS, EC, DCC, SV$
3. Test $\beta_{i,t}$ have different statistical properties across 75/25 EPU sub-samples.
4. For each $\beta_{i,t}$
 - 1 or 2 thresholds EPU^h, EPU^l ?
 - Difference in μ_β or σ_β ?

OUT-OF-SAMPLE: 2011-2022

1. OLS static β_{ols}
2. OLS conditional $\beta_{ols,t}$
3. Use $\beta_{ols,t}$ to properly set up the calculation of r_{hit} . See above discussion on Eq (6-7).
4. Formal evaluation of hedging effectiveness based on r_{hit} from state-depending strategy, against benchmarks: See Sections 4.1 and 4.2.
 - 4.1: Use EPU_t to guide the strategy switch (or rebalance) decision, depending on in-sample analysis of hedge ratios.
 - 4.2: EPU-adjusted OLS hedge ratios $\beta_{ols}^e, \beta_{ols}^{ed}$ against unadjusted hedge ratios $\{\beta_{ols}, \beta_{ec}, \beta_{dcc}, \beta_{sv}\}$